RANYA RAJESH PRASAD

Software Development Engineering · Java, Python, C++ · AWS · AI-Assisted Development · Ownership from Design to Operation

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EDUCATION

University of Illinois Urbana-Champaign — B.S. Electrical Engineering, Grainger College of Engineering **Expected May 2029**

Sophomore, **junior standing by credit hours** • GPA **3.65**/4.00 • **Minor: Hoeft Technology & Management Program** (joint Gies College of Business and Grainger College of Engineering) • Available Summer 2027

Coursework: ECE 220 Computer Systems & Programming (C, C++, assembly, memory), **ECE 120 Introduction to Computing (digital logic, computer organization)**, ECE 110 Introduction to Electronics, ECE 210 Analog Signal Processing, CS 173 (Discrete Structures), MATH 241 (Multivariable Calculus), MATH 257 (Linear Algebra) • CS 225 Data Structures in C++ (Spring 2027)

RECOGNITION

Corresponding author, peer-reviewed international publication, "SkinSight: An App Leveraging Neural Networks for Skin Cancer Detection," *Intl. Journal of Pharma Medicine & Biological Sciences* (2025) · **Oral presentation, ICBCB 2025** (IEEE co-sponsored, Seoul National University) · **First Place, shsHacks** (judged by Hudson River Trading) · Illinois State Scholar (top 10%)

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript / JavaScript, C, SQL • data structures & algorithms, object-oriented design

Backend & data: REST API design (Flask, Node.js / Express), PostgreSQL and MySQL (SQL), Firebase (NoSQL), React, Android

Cloud & DevOps: AWS (Cognito, ECS/Fargate, ECR, CloudWatch), Docker, Terraform, GitHub Actions CI/CD, Linux, Git, Agile

AI-assisted development: Claude Code, GitHub Copilot; I validate AI-generated code with review and tests before merge • pytest

EXPERIENCE

Jr. Applied AI Developer (IRAP) — *AbbVie* (contract through HireGenics) · Information Research **Oct 2026 – Present**

– Building **evaluation frameworks that decide when AI-generated output can be trusted**: golden test sets, LLM-as-judge scoring and regression detection, wired into **CI/CD with automated gates and rollback**.

Software Developer Intern — *Fylm TV, Inc.* · AWS, Terraform, Docker, CI/CD · distributed team **Aug 2026 – Present**

– **Cloud infrastructure**: own the Terraform module for the platform's authentication layer on AWS in a shared codebase; rebase and resolve conflicts against the team's baseline without breaking shared environments.

– **Quality gates**: hardened the pipeline that checks every pull request, with human approval before live changes.

– **AI software**: building the content-tagging service that derives descriptive tags from a film's video and audio tracks.

Software Developer — *NOBE* · client engagement for a registered investment advisory firm · Python, React **Sep 2026 – Present**

– **Financial software for a paying client**: building sub-account features where holdings, cost basis and ledger history are scoped per sub-account, and a return engine **validated against the client's 13-scenario workbook**. Delivered the first navigation workflow for client review.

LEADERSHIP & PROJECTS

Software Chair — *WinnieWheels autonomous delivery robot* · Women in ECE, UIUC · C++, Arduino, Raspberry Pi **Oct 2025 – Sep 2026**

– **Problem decomposition**: appointed as a first-year student, one of two software chairs **leading software for a 50-person team**; split the work into independent modules, set weekly goals, **ran code reviews**, and coordinated with the hardware chairs.

– **High-level design**: wrote the motor algorithm in Node.js; the obstacle-avoidance loop runs in **C++ firmware on the microcontroller** (two ultrasonic sensors in, a motor driver out), a control server coordinates, and a Raspberry Pi streams video, so the robot stays safe if Wi-Fi drops.

– **Debugging on real hardware**: a three-sample validity filter for noisy sensor readings; throttling and deduplication for a saturated serial link; a difference-threshold turn policy that ended left-right oscillation.

– **Testing without hardware**: built a simulator on the identical code path so every member could run the full stack. Taught the motor-control code to members; ran the robot live at Engineering Open House. Term ended Sep 2026; still a member.

SkinSight — AI software running on the device · *TensorFlow / Keras, TensorFlow Lite, Java (Android)* · Technical Lead **2023 – 2025**

– Trained and compared four neural networks over three trials each: ANN (31.19%), FFNN (64.20%), CNN (73.58%) and **Transfer Learning (84.69% mean, 90.32% best)**, significant vs. the CNN (t-test, p = 0.021). **Ran the chosen model on the phone itself** through TensorFlow Lite, with no server. Published and presented internationally.

Parallel computing: genome variant-calling pipeline · *Linux cluster on Google Cloud; bwa, samtools* · UChicago **Summer 2023**

– A head node dispatched jobs to worker nodes that aligned, genotyped and annotated in parallel, tuned by worker count and threads per node; **found the bottleneck stages** and scaled from one chromosome (2 hours) to all 23 (16 hours) at 8× the cost.

NetWatch — automated validation on every change · *Python, GitHub Actions, pytest* **2026**

– Log-triage agent that parses device logs and flags link flaps and login storms with statistical baselines plus rules. **CI scores every change against a labeled test set**, tracks latency and cost, and blocks the change if accuracy drops.

Cafe Queues — efficient, correct software under load · *React, Flask, PostgreSQL* **2022 – 2026**

– Campus pre-order site students used daily at peak lunch. Added a Flask REST API on a normalized PostgreSQL schema; a UNIQUE constraint and a transaction lock enforce one ticket per order: **20 simultaneous orders issue 20 distinct tickets**.

Also: Lead emcee, national STEM conferences (300–700 attendees) · EXPO Technical Lead, Engineering Council